Operator Packages, Proposer Strength, and Construction-Family Plateaus in Office-Scale Verified Search

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Abstract

Verified search, in which a language model proposes programs, a hard evaluator scores them, and selection keeps the best, has recently moved mathematical records; controlled ablations of the proposer-side components remain rare. We instrument a minimal FunSearch-style loop at office scale (a 30B local model on a laptop, 120–600 verified samples per run) with three operator packages: a schematic notebook the model writes and carries instead of verbatim elites, a named obstacle, and behavioural repulsion from constructions already found. On nine construction problems from a public repository, the complete 2^3 factorial with two replicates supports the primary contrast in a two-look sequence: the composition closes more of the seed-to-record gap ($+0.196$ $[+0.059, +0.343]$, $p = 0.023$; median per-problem effect $+0.045$). Repulsion raises functional diversity everywhere ($+0.31$, $p = 0.0039$; the operator partly optimizes this quantity). The factorial rejects anchoring as an interaction: the gain decomposes additively, the notebook package carrying the largest share, and memory+repulsion is the only arm that never collapses (0 of 18 runs) while matching the full composition to within 0.025. A frontier proposer under the identical loop reaches in tens of samples what the local model does not in hundreds, and its gains arrive without the operators. The search stalls after closing $\approx$92% of the gap toward the best published construction on the flagship problem, and the registered family-hint test gives the stall its first reading: named in words, the reference family is adopted and loses; handed as code, it is optimized but its grid ceiling lies below the plateau reached unaided. The loop transports and optimizes ideas; it did not originate one. We release the harness, every candidate, and the dated pre-registrations.

1 Introduction

FunSearch [13] and AlphaEvolve [10] demonstrated that a language model inside an evolutionary loop with a hard evaluator can produce genuinely new mathematical constructions. The systems that followed improved the *scaffolding*: sample-efficient archives and novelty rejection [6], verbal reflections as search gradients [16], stronger databases and routing. Two questions have stayed unmeasured. First, *which parts of what the proposer is asked to think* (its memory of the lineage, its statement of the obstacle, its relation to what was already found) change the search, and by how much? The prompts of the published systems show elite programs verbatim and vary everything else. Second, *how much of the outcome is the proposer itself*, holding the loop fixed? Published systems change model and scaffold together.

This paper measures both, at a scale any laboratory can afford, with the discipline our earlier studies [11] found necessary: every battery pre-registered before running, dated amendments, exact paired tests, and negative results reported at the same volume as positive ones.

The operators we test are not arbitrary: they are the three interventions that survived a two-paper program on open-ended generation: a *schematic memory* (the model's own compressed recap outperformed verbatim context on document-level integration), a *standing question*, and *repulsion* from the already-produced, which in weight space collapsed without an anchor and in behaviour space is the natural tail-preserving form.

Contributions. (1) A pre-registered program of cognitive operators inside a fixed verified-search loop, grown from a $5\times6\times2$ ablation to the complete $2^3\times9\times2$ factorial: the composition reaches +0.196 ($p = 0.023$ over the two-look sequence), behavioural repulsion buys functional diversity in every problem and replicate ($p = 0.0039$), and no single operator is safe alone. (2) A revision, by the completed factorial, of this program's recurring *anchoring pattern*: the memory×repulsion interaction on mean gap closed is zero: anchoring is not superadditive gain but *collapse prevention*. Repulsion without memory still breaks (three collapses in eighteen runs; four when paired with agenda), while memory+repulsion never does, echoing unanchored preference optimization in weight space. (3) A *scaling inversion*: from a 120-sample tie, the composed arm climbs to 0.3816 at 600 samples while repulsion alone stalls at 0.3514: diversity without accumulated direction does not compound. (4) A controlled *proposer comparison*: local 30B versus a frontier model under the identical loop and budget, on the same problem, locating the residual gap in the proposer rather than in the loop, with a frontier-strength scoping run in which the operators' mean contribution sits inside run-to-run noise. (5) An honest calibration of office-scale verified search, reaching 57–99.9% of the reference values (median 89%) at 120–600 samples, and a characterized wall: grid resolution, verifier time and proposer strength each unmoved within the tested settings, leaving the construction family as the leading suspect for the binding constraint.

2 Related work

Verified search systems. FunSearch evolved single functions with small code models and millions of samples [13]; AlphaEvolve evolves whole files with frontier ensembles and thousands [10], and its repository of 67 problems with official verifiers [3] is the substrate of our experiments. ShinkaEvolve reports record-level results with ∼150 samples using novelty-based rejection and bandit model selection [6]; ReEvo uses model-written reflections as "verbal gradients" [16]. Our schematic notebook differs from ReEvo's reflections in being a persistent, compressed lineage memory that *replaces* verbatim elites rather than commenting on pairs. A recent wave adds persistent memory to evolutionary code search: ε-MemEvo transfers compact strategy summaries across tasks with an adaptive injection gate and reports gate ablations [8]; EvoMem reuses mutation knowledge across runs [14]; MEMOIR shares algorithmic and failure-mode summaries across search branches [5]. What remains absent, and what we contribute, is a complete factorial, within one task suite and under a fixed local proposer, of a schematic-notebook package, an obstacle line, and an exact behavioural duplicate control. LoongFlow [15] composes planning, hybrid memory and MAP-Elites niches into one system and reports large efficiency gains; it is the closest architecture to our composed arm, and exactly the kind of system whose components an ablation like ours aims to isolate.

Why diversity needs help. Without selection pressure, LLM mutation chains collapse into attractor regions [4]; RL-tuned models lose pass@k coverage relative to their base [17]; and coverage under repeated sampling grows log-linearly only when a verifier exists [2]. Our behavioural-repulsion operator is novelty search [7, 9] transplanted into the proposer's prompt, with the construction itself (not an embedding) as the behaviour descriptor.

3 The harness, the operators, the problems

Loop. A minimal FunSearch-style loop: K islands, each holding its best programs; per generation, per island, a prompt is built and S candidates sampled; each candidate runs in a sandbox (time and memory limits) and is scored by the problem's verifier; islands keep bounded elites and migrate their best every third generation. The proposer is `Qwen3-Coder-30B-A3B-Instruct` (8-bit, MLX, on a MacBook M5 Pro) prompted through its chat template; Section 5 swaps in Claude Opus 5 through an API under the identical loop. Every candidate is logged: code, score, error, generation, island, and a behaviour signature (a hash of the rounded construction it outputs).

Operators. **A** (baseline): the prompt shows the statement and the island's top-3 programs with scores, FunSearch-style. **B** (schematic notebook): each generation the model writes a $\leq$12-line notebook (what worked, what failed and why, the open question) from the elites and the recent failures; the prompt then shows the notebook and only the single best program. **C** (agenda): one model-written line naming the structural obstacle of the current best, appended to the prompt. **D** (behavioural repulsion): candidates whose construction duplicates an island member's are rejected, and the prompt lists already-found constructions as forbidden. **E**: B+C+D. We refer to these as operator *packages*: each bundles several changes at once (B replaces three verbatim elites with one, adds failure summaries and an open question, and spends an extra call; D combines a prompt-level forbidden list with a post-verification duplicate filter), so the factorial isolates packages, not single cognitive mechanisms; a finer within-package decomposition is registered as future work.

Problems. Nine construction problems ported from the official repository notebooks, chosen for cheap exact verification and known headroom. The original battery: circle packing (26 circles, maximize the sum of radii), ring loading (m=15, exact 2^m verification), beat-the-average (a pmf maximizing $P[X_1+X_2+X_3 < 2X_4]$), the first autocorrelation inequality (minimize an upper bound on C_1), isosceles-free subsets of the 64^2 grid, and sum-difference III. A pre-registered extension, run before any n=9 analysis, added three: the max–min pairwise-distance ratio for 16 points in the plane (minimize), the Heilbronn triangle problem for 11 points (maximize the smallest triple area), and splitting 180! into 180 factors maximizing the smallest (exact big-integer verification). Each run's budget is 120 verified samples (10 generations $\times$ 2 islands $\times$ 6); the primary metric *frac* is the fraction of the seed-to-record gap closed, *auc* its mean over the budget (sample efficiency), *div* the share of distinct behaviours among valid candidates.

4 The factorial

Pre-registered contrasts (exact paired sign-flip over problems, cells = problems, pooled replicates). The original six-problem battery put the primary contrast at the design's resolution floor (+0.199, p = 0.0625; with six cells and one adverse problem the exact test cannot go below 4/64); the prospectively registered nine-problem extension supports it: **F1** (primary), E vs A on frac: +0.196 [+0.059, +0.343], $p = 0.023$ over the combined nine problems. We report the stages separately, since the combination is a two-look sequence rather than an independent confirmation: the three extension problems alone are positive but underpowered (+0.19 on average, $p = 0.25$, carried mostly by max–min distance). The effect is also heterogeneous (Figure 1): the median per-problem effect is +0.045, leave-one-problem-out means span +0.153 to +0.224, with large gains on four problems, little on four, and one small loss. **F4**, D vs A on functional diversity: +0.307 [+0.149, +0.477], $p = 0.0039$, the minimum attainable p, positive in all nine problems in both replicates. **F6** (the anchor), E vs D: +0.137 [-0.047, +0.382], $p = 0.156$: not

frac (pooled over 2 replicates)	A	B	C	D	BC	BD	CD	E
circle packing 26	0.703	0.974	0.976	0.000	0.989	0.752	0.980	**0.992**
ring loading 15	0.444	0.767	**0.808**	**0.808**	0.731	**0.808**	**0.808**	**0.808**
beat-the-average	0.181	0.384	0.519	0.608	0.661	**0.789**	0.540	0.709
autocorrelation C_1	0.059	0.156	0.041	0.202	0.014	**0.212**	0.000	0.104
isosceles-free 64^2	**0.560**	**0.560**	**0.560**	0.399	**0.560**	**0.560**	**0.560**	**0.560**
sum-difference III	**0.392**	0.329	0.253	0.158	0.314	0.327	0.069	0.362
max–min distance 16	0.169	0.992	0.372	**0.993**	0.919	0.793	0.247	0.708
Heilbronn triangle 11	0.672	0.431	0.507	0.540	0.371	0.477	0.619	**0.700**
180! into 180 factors	**0.774**	**0.774**	**0.774**	**0.774**	**0.774**	**0.774**	**0.774**	**0.774**
mean frac	0.439	0.596	0.534	0.498	0.592	0.610	0.511	**0.635**
mean functional diversity	0.41	0.53	0.52	0.72	0.59	0.68	**0.80**	0.70
collapses (frac < 0.05, of 18 runs)	2	1	1	3	2	**0**	4	1

Table 1: The complete 2^3 factorial over nine problems, pooled over two independent replicates (120 verified samples per run). frac = fraction of the seed-to-record gap closed. E has the best mean and is best or tied-best in five of nine problems; D alone is bimodal: best on wide valid manifolds (pmfs, sequences), catastrophic where validity is fragile (packing geometry, constrained integer sets); BD is the only arm that never collapses, CD the one that collapses most; two problems (isosceles-free, 180!) are family plateaus where every arm meets the same wall.

supported as a pairwise contrast; the factorial below resolves what it was measuring. F2 (auc): +0.115, $p = 0.070$: the composed arm's advantage builds late. (The operators' measured overhead is small: one extra call per island-generation per operator, ∼220 characters each, 1–3% of proposer output; Section 8 gives the accounting.)

The completed 2^3. Three pairwise arms (memory+agenda, memory+repulsion, agenda+repulsion) over the same nine problems and two replicates complete the cube (8 cells $\times 9 \times 2$; pre-registered before running). The effect-coded analysis (exact sign-flip over problems, 2^9) rejects the primary interaction: memory×repulsion $\beta = +0.003$ [-0.039, +0.035], $p = 0.47$ one-sided; agenda×repulsion is null (-0.007, $p = 0.79$); and no main effect survives BH correction: schematic memory is the largest single term ($\beta = +0.057$ $[+0.014, +0.105]$, raw $p = 0.047$, $q = 0.14$), agenda (+0.016) and repulsion (+0.011) near zero on frac. The decomposition is exact ($E - A = 2\,(\beta_M + \beta_A + \beta_R + \beta_{MAR}) = 0.196$): the certified composite gain splits as roughly 58% memory, 16% agenda, 12% repulsion and 14% three-way (a descriptive decomposition with wide per-term intervals); at $n = 9$ the total certifies, the parts do not. Two patterns in the cube are worth the trip. First, each operator has one job (Table 1, Figure 2): the four memory-on cells hold the four best mean fracs, while every repulsion-on cell sits at 0.68–0.80 functional diversity against 0.41–0.59 without it; directionally, memory moves the gap and repulsion moves diversity, with the caveat that the memory main effect does not survive correction. Second, the anchor lives in the tail, not the mean: memory+repulsion is the only arm that never collapses (0 of 18 runs, against 3 for repulsion alone and 4 for agenda+repulsion, the cube's worst), so what memory buys repulsion is insurance against starving the island, not superadditive mean gain. Agenda, decomposed at last, pays for neither: near-zero on frac alone, and the worst collapse count when paired with repulsion. Exploratory pairwise contrasts back the parsimony reading: E vs BD +0.025 ($p = 0.55$, median 0.000), E vs B +0.039 ($p = 0.47$), BD vs B +0.014 ($p = 0.81$), all null at $n = 9$. The collapse asymmetry is robust to the threshold: at frac ≤ 0, < 0.01, < 0.05 and < 0.10 the counts stay 3/3/3/3 for repulsion alone and 3/4/4/4 for agenda+repulsion against 0/0/0/0 for memory+repulsion, and no run anywhere produced zero valid candidates, so these are performance collapses, not validity collapses.

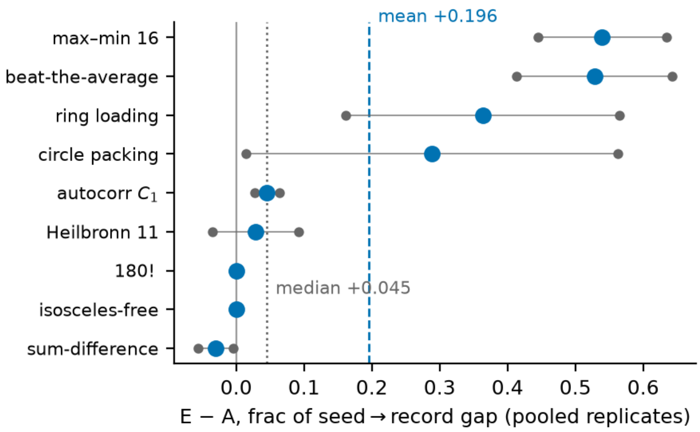

Figure 1: The primary contrast, problem by problem: E − A on the fraction of the seed-to-record gap closed (pooled over two replicates). Small grey dots are the two replicates (lines join them); the large dot is their mean. The pooled mean is +0.196; the median, +0.045; leave-one-problem-out means stay between +0.153 and +0.224. Within-problem replicate spread is large on several problems (circle packing: +0.01 and +0.56), which the problem-level test does not see.

The heterogeneity is the finding the factorial adds to the literature: *where* the valid manifold is wide (a pmf, a sequence of pairs), repulsion alone is the best single operator; where validity is fragile (non-overlapping circles, exact integer constraints), it starves the island (both replicates of circle packing collapse), and the composed arm inherits most of the benefit with none of the catastrophes. No single package avoids every failure mode; memory+repulsion was the only configuration that avoided all of them in these runs.

5 Scale, proposers, and the wall

We then took the problem with guaranteed headroom, beat-the-average, where AlphaEvolve's own reported value (0.3890) sits below the best known (0.400695), and ran a pre-registered ladder. (i) *Budget*: at 600 samples the composed arm reaches 0.3816 while repulsion alone, which had tied it at 120 samples, stalls at 0.3514: diversity without accumulated direction does not compound. (The same 5× extension on max–min distance 16 left both local arms *worse* than their own 120-sample runs on a fresh seed: seed variance dominates budget there too.) (ii) *Grid*: refining the pmf grid from $L{=}5000$ to $L{=}20000$ at equal budget makes the local proposer *worse* (0.3561): the limit was not resolution. (iii) *Proposer*: Claude Opus 5 under the identical loop reaches 0.3897 within ∼100 samples (just above the repository's published value, obtained with 2025-era proposers; we read this as locating the wall, not as ranking systems across model generations) and plateaus there for 17 generations. (iv) *Verifier time*: tripling the sandbox budget to 60 seconds, so heavier optimizers can run, moves the plateau by +0.0002.

The constructions are legible. The local model's best is a bimodal sparse measure of 16 atoms (half the mass on $\{0, 1, 2, 3\}$, geometrically spaced atoms at the top), with the strategy stated in comments. Claude's best implements an exact sparse-atom objective with an analytic gradient and a multiplicative-weights refinement. The frontier proposer and AlphaEvolve's published value sit within 0.0007 of each other; the local model stalls about 0.008 behind them. All three lie at 92–98% of the seed-to-reference gap toward a reference whose construction evidently lies outside the sparse-atom family every language model proposes. More precisely: the candidates of both proposers are small finitely supported measures found numerically, while the Bellec and Fritz [1] construction is a multiscale, recursively structured sequence of discrete measures whose analysis shows a finitely supported measure cannot be optimal; the family-hint conditions below

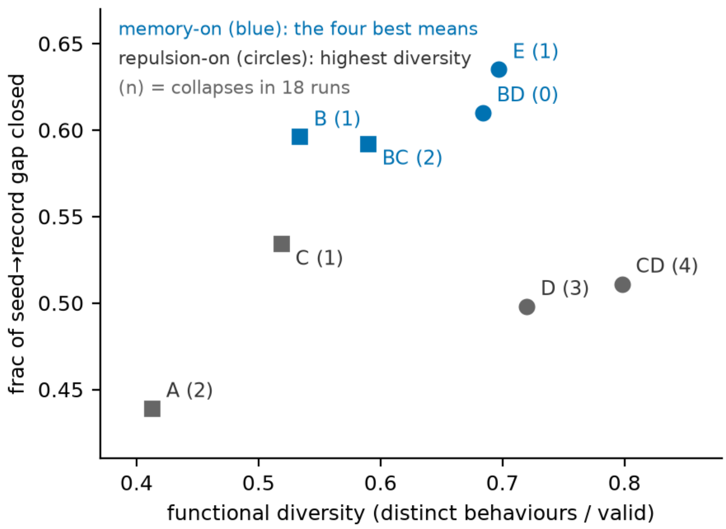

Figure 2: The completed 2^3 at a glance (cell means over nine problems, two replicates). Memory-on cells (blue) hold the four best gap fractions; repulsion-on cells (circles) hold the highest functional diversity; the two axes barely interact ($\beta_{M\times R} \approx 0$). Collapse counts in parentheses: memory+repulsion is the only arm that never collapses; agenda+repulsion collapses most.

name that structure. Within the settings tested, grid, time and proposer strength are ruled out; what remains is the family: an idea.

A second problem sharpens where such ideas live. On max–min distance 16, the frontier proposer under the same loop emits *in generation zero*, before any search feedback, a minimax program (SLSQP on $\min t$ s.t. $d_{ij}^2 \leq t$, $d_{ij}^2 \geq 1$, analytic Jacobians) whose value lies marginally below the repository's published one; every valid sample across the run converges to the same value. The loop verified this construction; it did not discover it. We treat the datum as scope, not as a record claim: the operators structure search and protect its tails, but they do not inject construction knowledge the proposer lacks. The pre-registered scoping runs make the same point from the other side, on all three registered problems. On beat-the-average, the bare, memory-only and repulsion-only arms land within 0.001 of one another (0.3871, 0.3866, 0.3876; single runs, estimation only) and within 0.003 of the composed arm's 0.3897. On max–min distance and on the Heilbronn problem every arm converges to an identical value (12.889230 and 0.036530 respectively; the Heilbronn reference in our registry floors the reported figure, so that value reaches the frozen reference and supports no exceedance claim). At frontier strength the operators' mean contribution sits inside run-to-run noise everywhere we measured it: the certified gains are a local-proposer phenomenon, and whether the collapse insurance survives at this strength remains untested.

6 Discussion

What office scale buys. With 120–600 verified samples, a local model and one laptop: 57–99.9% of the best known values (median 89%) across nine problems, legible programs, and, with a frontier proposer at ~US$50 of API, parity with the values the large systems report on this problem. The published record-movers used frontier ensembles with thousands of samples [10], millions with small models [13], or open-source loops with strong API models [6]; the band we measure (local proposer, hundreds of samples, controlled operators) was unoccupied, and it is the band most laboratories can afford.

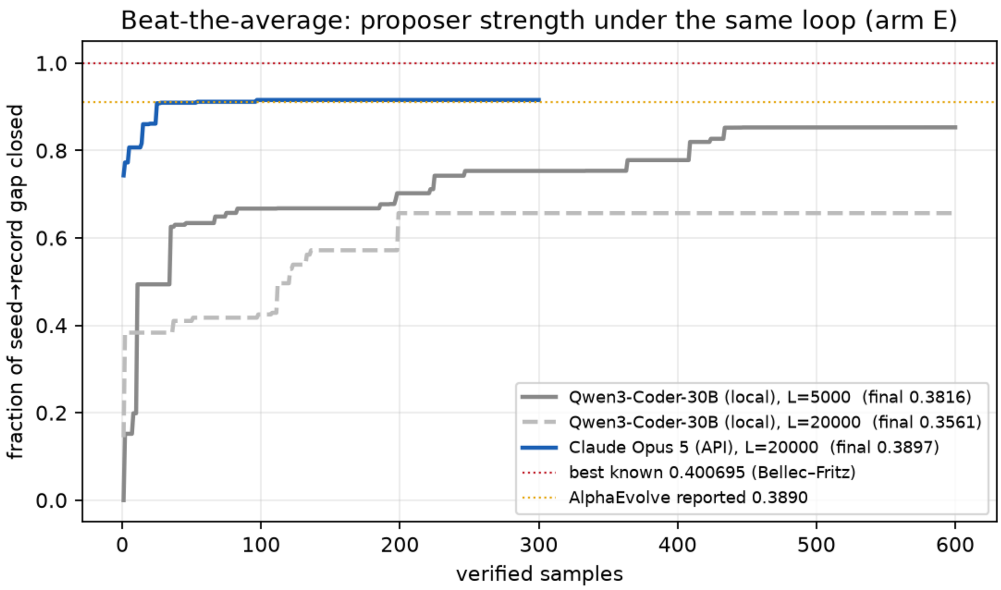

Figure 3: Beat-the-average under the same loop (arm E): fraction of the seed-to-record gap closed per verified sample. The local 30B at two grid resolutions; Claude Opus 5; dotted lines mark AlphaEvolve's reported value (0.3890) and the best known (0.400695, Bellec–Fritz).

The operator ledger. E had the highest mean and one collapse; memory+repulsion was the only zero-collapse arm, at a mean within 0.025 of E's and without the agenda's extra call, so the parsimonious operational recipe these data support is memory+repulsion, with the full composition as the highest-mean variant. The mechanism is visible in the notebooks: the schematic memory keeps direction across generations and the repulsion spends the saved budget on genuinely different constructions. Alone, each fails somewhere: memory and agenda are inert on problems the model already saturates; repulsion starves fragile manifolds. This is a recurring collapse-prevention pattern, consistent with the anchoring hypothesis, that this program has now observed across narrative generation, weight-space preference optimization, and prompt-space search.

The proposer comparison and the plateau. On this one problem, the frontier proposer is far more efficient in verifier evaluations (+0.008 absolute over the local model's 600-sample best, reached with roughly 20× fewer verified samples), and everything we tried after it moved little (grid, time, more samples: +0.0002). This is a sample-efficiency comparison on one problem under one harness, not yet a proposer-strength curve; replications on further problems are planned, and reasoning effort, ensembles, alternative representations and thousand-sample budgets all remain untested explanations. The shared plateau suggests, consistently with our consolidation results [12], that search over a fixed prior rapidly collects what the prior's construction families contain and then stalls at the best family member, a hypothesis to test rather than assert. *The family-hint experiment* (registered before running; five conditions × three seeds, 120 samples, memory+repulsion arm) put it to the test on beat-the-average: (a) no hint; (b) a generic hint; (c) a structural hint naming the Bellec and Fritz [1] family without parameters; (d) a crude in-family seed program (score 0.2734); (e) our best hand instantiation of the family as an oracle seed (0.3742; on this grid with strict inequality, notably *below* the 0.3816 sparse-atom plateau, a fact measured and registered before launch). The answer came with a twist. Condition means over three seeds: no hint 0.352, generic 0.333, structural 0.327 (the worst), family seed 0.358, oracle 0.375. The structural hint *worked as an instruction and lost as a search*: its final constructions carry the family's signature (an atom at zero, atoms accumulating toward the maximum across four log-distance bands), so the model obeyed, entered a family that is inferior at this grid, and paid about 0.025 for it. Handed the family as code, the loop optimizes it well (+0.085 from the crude seed, above the no-hint endpoint in three of three seeds) and stays loyal to it (no oracle run deserted to the better sparse family). No unaided run proposed the family. Within this problem and grid, then, the registered hypothesis was *not* supported: handing the idea, in words or in code, did not cross the last mile, and the idea itself was not the missing step at L=5000. What the experiment does establish is a division of labour: the loop transports and optimizes ideas; it does not originate them, and instructing one is not inducing one, the third appearance of that pattern in this program (the continuity connective of the companion study; the frontier scoping runs; now the structural hint). This is the question Georgiev et al. [3] report answering with an expert in the loop, from the other side.

Design rules. For verified search at small scale: pair memory with repulsion (the zero-collapse arm; the full composition adds +0.025 of mean at the cost of the agenda's extra calls, a difference our data cannot distinguish from zero), and never run repulsion alone on fragile manifolds; keep the behaviour descriptor exact (the construction, not an embedding); spend on the proposer before spending on samples; treat plateaus as candidate family boundaries and test them (grid, time, proposer and hints in pre-registered steps) rather than declaring them.

7 Experimental provenance

Every battery was registered before running; the combined analyses are sequential, and the table makes the order of looks explicit.

battery	registered	data already seen	primary	status
F (5×6×2)	2026-08-22	none	E>A frac	$p = 0.0625$ (floor)
F-ext ladder	2026-08-23	F	budget/grid/proposer	descriptive
F-ext2 (n=9)	2026-08-24	F, ladder	E>A frac	$p = 0.023$ combined
2^3 completion	2026-08-24	F, ext2 partial	M×R > 0	rejected ($p = 0.47$)
Frontier scoping	2026-08-26	all above	estimation only	operators $\approx$ null
Family hint	2026-08-27	all above	optimize-once-given	yes as code, no as words

Table 2: Order of registrations and looks. The nine-problem primary analysis is a two-look sequence (discovery + prospective extension), not an independent confirmation.

8 Limitations

Nine problems is still a small universe of construction families; the full 2^3 isolates interactions, but at n=9 no per-term effect survives correction, so the operator decomposition (memory largest) is directional, not certified, and the collapse asymmetry is descriptive; budgets are equalized in proposals and verifier evaluations (120 of each per run; duplicate rejection in the repulsion arms happens after verification, since the signature requires the construction, and discards 20–30 verified candidates per run from the islands); the operators' extra calls are measured at one per island-generation per operator ($\sim$220 characters per call, an overhead of 1–3% of proposer output characters), so sample efficiency here means verifier-evaluation efficiency at nearly matched token budgets; the diversity metric is related to the repulsion operator's own filter; the behaviour signature is a hash of the returned construction after rounding, with no canonicalization for symmetries (permutation, rotation, reflection), so div can count symmetric variants as distinct, its rounding sensitivity is untested, and the D filter and the div metric share this signature; the six problems satisfy criteria fixed before any runs (cheap exact verifier, short-function candidates, known headroom; the universe of 68 candidate problems and per-problem ratings are in the released reconnaissance table) but were not sampled randomly; *frac* is floored at the seed by construction, signs are harmonized for minimization, and reference values are frozen at the repository commit recorded in the release; the scaling ladder is one domain; the API proposer is not bitwise reproducible and its reasoning effort was fixed low; comparisons against repository values compare a 2026 proposer with values obtained by 2025-era systems, so they locate our wall and never rank systems: no system-level head-to-head with ShinkaEvolve or AlphaEvolve is attempted, and the operators should be read as structuring search, not as a substitute for construction knowledge the proposer lacks.

9 Conclusion

Inside a fixed verified-search loop at office scale, the operator packages our generation studies isolated (a schematic notebook, a named obstacle, behavioural duplicate control) compose into the arm with the highest mean, E, at one collapse in eighteen runs; memory+repulsion was the only zero-collapse configuration, within 0.025 of E's mean, and is the parsimonious recipe these data support. Repulsion alone enforces behavioural non-duplication by construction and was the best single package exactly where the valid manifold is wide; and on the one problem where we compared them, a frontier proposer reached a higher value with roughly 20× fewer verifier evaluations. On beat-the-average both proposers stalled inside one small finite-support construction family, below a multiscale reference; the other plateaus are consistent with, but do not establish, analogous family boundaries. Whether crossing that boundary requires what we have informally called an idea was put to a registered test on the flagship problem: the idea, handed in words, was followed and lost; handed as code, it was optimized but its grid ceiling lies below the plateau the models reach unaided. The loop moved ideas; it did not originate one, and the one we handed it was not the missing mile here. Whether the last mile of these searches is an idea, the question this program has carried informally, remains open, now with its first measured answer.

References

[1] Pierre C. Bellec and Tobias Fritz. Optimizing over iid distributions and the beat the average game, 2024. v5, 2025.

[2] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling, 2024.

[3] Bogdan Georgiev, Javier Gómez-Serrano, Terence Tao, and Adam Zsolt Wagner. Mathematical exploration and discovery at scale, 2025.

[4] Can Gurkan, Forrest Stonedahl, and Uri Wilensky. Mutation without variation: Convergence dynamics in LLM-driven program evolution. In *GECCO Companion (Workshop on LLMs for/with Evolutionary Computation)*, 2026. arXiv:2606.05408.

[5] Fatemeh Haji, Javier Delarosa Quiros, and Peyman Najafirad. Memory-guided tree search with cross-branch knowledge transfer for llm solver synthesis, 2026.

[6] Robert Tjarko Lange, Yuki Imajuku, and Edoardo Cetin. Shinkaevolve: Towards open-ended and sample-efficient program evolution, 2025.

[7] Joel Lehman and Kenneth O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.

[8] Aofan Liu, Shiyuan Song, and Yiyan Qi. ε-memevo: Adaptive cross-task memory transfer for llm program evolution, 2026.

[9] Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites, 2015.

[10] Alexander Novikov et al. Alphaevolve: A coding agent for scientific and algorithmic discovery, 2025. Google DeepMind; arXiv:2506.13131.

[11] Roberto I. Ono Filho. Interrupting the loop: Periodic subject changes raise judged surprise and connection in base language models, 2026.

[12] Roberto I. Ono Filho. Moving the mean, not the best: What inference-time interventions and weight consolidation buy in open-ended generation. Manuscript, under revision; to appear on arXiv, 2026.

[13] Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, Pushmeet Kohli, and Alhussein Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024. doi: 10.1038/s41586-023-06924-6.

[14] Viktor Volkov, Valentin Khrulkov, Andrey V. Galichin, Danil Sivtsov, Nikita Glazkov, Olga Volkova, Konstantin Pchelin, Iaroslav Bespalov, Dmitry V. Dylov, Petr Anokhin, and Ivan Oseledets. Evomem: Memory-augmented evolution for code optimization, 2026.

[15] Chunhui Wan et al. Loongflow: Directed evolutionary search via a cognitive plan-execute-summarize paradigm, 2025.

[16] Haoran Ye, Jiarui Wang, Zhiguang Cao, Federico Berto, Chuanbo Hua, Haeyeon Kim, Jinkyoo Park, and Guojie Song. Reevo: Large language models as hyper-heuristics with reflective evolution. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2402.01145.

[17] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems*, 2025. arXiv:2504.13837.